

FABDEM-ML pipeline: data → features → spatial-CV training → outputs + OOD test

Data sources

FABDEM v1.2
(GEE asset)

ICESat-2 ATL08 v7
(earthaccess)

Sentinel-1/2
(Planetary Computer)

Per-tile pipeline

Terrain features
• SurtGis (Rust)
• 17 terrain + 8 hydro

Footprint extraction
• Quality filters
• EGM2008 correction

Satellite indices
• NDVI/NDWI/NDMI/BSI
• Sentinel-1 VV/VH dB

Feature stack
33 features
× 135,350 footprints

ML training

XGBoost regression
• Target: residual
($h_{te_ortho} - fabdem$)
• Optuna 100 trials TPE

Spatial-block CV
• 10-km blocks (n=570)
• GroupKFold K=5

OOD test
Apply to humid
temperate w/o retrain

Outputs
• Corrected DEM COG
• SHAP importance
• Per-tile metrics

